RESEARCH SUMMARY

1. Objective of the Study

This study addresses a critical gap in population-level diabetes screening: the absence of a clinically aligned, reproducible, and deployable machine learning system that prioritizes the identification of diabetic individuals over the optimization of statistical benchmark metrics. Existing approaches systematically underperform in clinical contexts because they optimize for overall accuracy at arbitrary default classification thresholds, apply inadequate or single-layer strategies against the class imbalance inherent in real-world diabetes prevalence data, and report performance using metrics that reward majority-class dominance rather than minority-class sensitivity.

The overarching objective of this project is to design, train, validate, and deploy a Smart Diabetes Monitoring and Prediction System on the BRFSS 2015 behavioral health survey dataset that achieves a binding minimum clinical Recall of 0.90, operates without laboratory access or clinical infrastructure, and is packaged in a form accessible to healthcare practitioners without requiring programming expertise. The system is intended to function as a population-level screening support tool that flags individuals warranting further clinical evaluation, not as a standalone diagnostic instrument [1], [5], [9].

2. Research Methodology

The study employs a supervised binary machine learning classification methodology applied to a large-scale behavioral health survey dataset. The complete pipeline spans eighteen sequential steps organized into four progressive experimental phases.

The BRFSS 2015 dataset, comprising 253,680 records, 21 self-reported behavioral and demographic features, and a binary diabetes target variable, was preprocessed through duplicate removal, IQR-based outlier capping applied to the seven continuous features, stratified 80/20 train-test splitting preserving the 6.18 to 1 class imbalance ratio, and RobustScaler normalization fitted exclusively on the training set to prevent data leakage [12], [25], [36].

Domain-informed feature engineering expanded the original feature representation from 21 to 45 attributes by introducing 24 clinically derived variables including WHO-standard BMI category encodings, composite cardiovascular and metabolic risk scores, a lifestyle quality index, a healthcare access index, a normalized health burden index, pairwise interaction terms, polynomial expansions of continuous predictors, and socioeconomic proxy variables [3], [37]. A consensus feature selection procedure combining Random Forest Mean Decrease in Impurity and Mutual

Information scores identified the top 15 most discriminative features for Phase 3 training [17], [38].

Seven classification algorithms were evaluated across three feature phases under a three-layer imbalance handling strategy combining algorithm-level cost-sensitive weighting through class_weight and scale_pos_weight parameterization, and evaluation-level clinically appropriate metric prioritization [9], [22]. Model selection was governed by a composite scoring function assigning 40 percent weight to Recall, 25 percent to F1-Score, 20 percent to ROC-AUC, 10 percent to MCC, and 5 percent to Balanced Accuracy. The globally best model, LightGBM from Phase 2, underwent hyperparameter optimization through RandomizedSearchCV with n_iter=50, Stratified 5-Fold cross-validation, and Recall as the optimization criterion [15], [42]. A systematic threshold optimization procedure evaluated 90 candidate thresholds from 0.05 to 0.95 in increments of 0.01, selecting the highest threshold satisfying Recall greater than or equal to 0.90 [33].

3. Sample and Research Instrument

The study uses the preprocessed version of the 2015 Behavioral Risk Factor Surveillance System survey, an annual population-based telephone health survey administered by the United States Centers for Disease Control and Prevention to adults across all fifty states [12]. The preprocessed version used in this study was prepared by Teboul and is publicly available through Kaggle [13].

The dataset contains 253,680 complete records with no missing values. The 21 input features span fourteen binary health and behavioral indicators including high blood pressure, high cholesterol, smoking status, physical activity, fruit and vegetable consumption, stroke history, and heart disease history, and seven ordinal variables including BMI, general health self-rating, mental and physical health burden in days, age category, education level, and income level. The binary target variable distinguishes diabetic or prediabetic individuals from non-diabetic individuals, with 35,346 positive cases representing 13.9 percent of the total sample and 218,334 negative cases representing 86.1 percent [12], [13].

The dataset was divided into a training set of 183,579 records and a held-out test set of 45,895 records using stratified splitting. The test set contains 7,019 diabetic cases and 38,876 non-diabetic cases, preserving the population-level class distribution for unbiased final evaluation [41].

4. Key Findings of the Study

The study produced seven principal findings.

First, LightGBM consistently outperformed all six competing algorithms across all three experimental feature phases when evaluated using the clinically weighted composite scoring function, achieving the globally highest composite score of 0.6944 in Phase 2. Its superiority is attributable to its leaf-wise tree growth strategy, its native cost-sensitive imbalance handling through scale_pos_weight, and its architectural efficiency advantages over standard gradient boosting [15], [19].

Second, domain-informed feature engineering from 21 to 45 features produced measurable improvements in composite performance across all seven algorithms, confirming that clinical domain knowledge systematically embedded in derived features provides predictive signal that raw survey items alone cannot capture. The interaction terms GenHlth_x_BMI and Age_x_BMI, alongside the CompositeRiskScore and CardioRisk composite features, ranked among the top predictors in all importance analyses [37], [17].

Third, hyperparameter optimization identified a configuration using 400 boosting trees, learning rate 0.05, maximum depth 5, 63 leaves, and scale_pos_weight 12, achieving a mean cross-validated Recall of 0.9095 across five stratified folds before any threshold adjustment, confirming that the clinical constraint is achievable at the model level alone [15], [42].

Fourth, threshold optimization demonstrated that the default classification threshold of 0.50 is clinically suboptimal for deployment, and that the threshold of 0.53 represents the optimal operating point satisfying the binding Recall constraint of 0.90 while maximizing precision within that constraint [30], [33].

Fifth, at deployment threshold 0.53, the final model achieves a Recall of 0.9064, correctly identifying 6,362 out of 7,019 diabetic cases in the held-out test set, with 657 false negatives, a ROC-AUC of 0.8186, PR-AUC of 0.4447, Balanced Accuracy of 0.7228, and MCC of 0.3114 [9], [43], [44].

Sixth, feature importance rankings demonstrate strong alignment between the model's learned predictors and established epidemiological evidence for Type 2 diabetes risk factors, providing face validity for clinical deployment. General health self-rating emerged as a standalone top predictor, suggesting that subjective health perception may precede formal clinical identification by years [7], [12].

Seventh, the comparison with seven representative prior studies on the BRFSS dataset confirmed that no existing published work simultaneously enforces an explicit minimum Recall deployment criterion, performs systematic threshold optimization, and applies a multi-layer imbalance handling strategy within a single reproducible pipeline [8], [27], [34].

5. Research Implications and Key Recommendations

The findings of this study carry four principal implications for clinical practice, public health policy, and future research in medical machine learning.

For clinical practice, the study demonstrates that a population-level diabetes screening tool operating entirely on self-reported behavioral and demographic data can achieve a sensitivity of 90.6 percent, a level approaching the clinical adequacy threshold for a first-line screening instrument. Deployment of such a system in primary care, community health, and telehealth settings could substantially reduce the proportion of undiagnosed diabetic individuals by identifying high-risk individuals who would not otherwise seek or access laboratory-based screening [5], [6], [1].

For public health policy, the cost-accessibility profile of this approach is particularly relevant for low- and middle-income populations where laboratory-based screening is financially or logistically inaccessible. A web-based screening tool requiring only a structured questionnaire could be integrated into existing community health worker programs, telehealth platforms, and population health management systems at minimal incremental cost [8], [5].

For the machine learning research community, this study provides concrete empirical evidence that the choice of decision threshold is as consequential as the choice of algorithm for clinical deployment, and that optimizing for recall-constrained threshold selection rather than accuracy or F1-Score alone produces substantially different and more clinically appropriate system behavior [30], [33]. The clinically weighted composite scoring function introduced in this project provides a reusable template for model selection in any imbalanced medical binary classification task.

The key recommendations arising from this study are:

- All future diabetes prediction systems reporting clinical deployment performance specify the decision threshold at which metrics are reported and justify it clinically rather than adopting the statistical default.
- External validation on geographically and demographically independent datasets be treated as a mandatory pre-deployment requirement rather than an optional extension.
- Longitudinal behavioral health survey data be prioritized as a future data source to enable trajectory-based risk modeling; and that formal clinical evaluation with real patient cohorts and clinician usability assessment be undertaken before any system of this type is considered for regulated healthcare deployment [34], [7], [5].